

Kinematic Analysis of Internal Mechanisms of Beginning Movement Load Theory Equipment

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Abstract

This study aims to analyze the dynamics of the internal mechanism of the beginning movement load theory equipment and identify the reasons for the variations in the load applied to the pulling parts during usage. To achieve this goal, we studied the beginning movement load theory, examined several patents of beginning movement load training equipment, and conducted a mechanical analysis of the internal mechanisms of the equipment.

Beginning movement load theory aims to minimize the load at the beginning of the movement, extend in a relaxed state, and contract with force. The main feature of this patent is the three-dimensional trajectory of the device, where the upward and downward loads increase the torque through the rotation of the handle through a gear set and a slider-crank mechanism. The dynamic analysis of the slider-crank mechanism shows that it not only generates opposing torque but also produces vertical and horizontal loads when the handle is pulled. We also used MATLAB to simulate trend charts, finding that the rotational angle and speed of the handle significantly affect the vertical load, potentially increasing it by about 1.5 times the weight. While the angular velocity has a minor impact on the trend of load torque changes, the torque variation follows a sinusoidal pattern with changes in the handle's rotational angle.

Keywords: Beginning movement load theory, mechanism, kinematic analysis